STUDY NOTES · EDITION 1.0

Generative Artificial Intelligence

An introduction with diagrams, core concepts and notes

What this guide covers

- Foundations: artificial intelligence, machine learning and deep learning
- Generative models: GANs, VAEs, diffusion models and transformers
- How large language models generate text, token by token
- Training, fine-tuning, prompting and retrieval-augmented generation
- Applications, risks and responsible use
- Summary, glossary and key research papers

Level: introductory · Suitable for BCA and other computing courses

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HOW TO USE THESE NOTES

Read the chapters in order. Each one builds on the terms introduced before it.

Figures are referenced in the text. Look at each diagram before reading its explanation.

Shaded **Key Notes** summarise what to remember. The **Glossary** at the end defines the main terms.

1 Introduction

1.1 Why Generative AI matters

- Generative AI is one of the fastest-growing areas of computer science. Tools that write essays, draw pictures, generate code and summarise documents now appear in everyday work and study.
- For a BCA student the subject connects several strands of the course: programming, probability and statistics, databases and cloud computing. Generative models are built with all of them.

1.2 Scope of these notes

These notes explain how artificial intelligence, machine learning and deep learning are related, what makes a model *generative*, and how the four main families of generative models work: **GANs**, **VAEs**, **diffusion models** and **transformer-based language models**. They then cover applications, risks and a glossary of key terms.

1.3 Learning objectives

- Explain how AI, machine learning, deep learning and generative AI are related.
- Distinguish discriminative models from generative models.
- Describe how GANs, VAEs, diffusion models and transformer-based LLMs produce new content.
- Explain tokens, embeddings, self-attention and sampling at a conceptual level.
- Identify applications, limitations and responsible-use concerns.

2 AI, Machine Learning and Deep Learning

2.1 The hierarchy

The four terms are nested. **Artificial intelligence** is the broad goal of building systems that perform tasks requiring intelligence. **Machine learning** is a subset in which the system learns patterns from data instead of following hand-written rules. **Deep learning** is a subset of machine learning that uses neural networks with many layers. **Generative AI** is a class of models, usually deep learning models, whose output is new content.

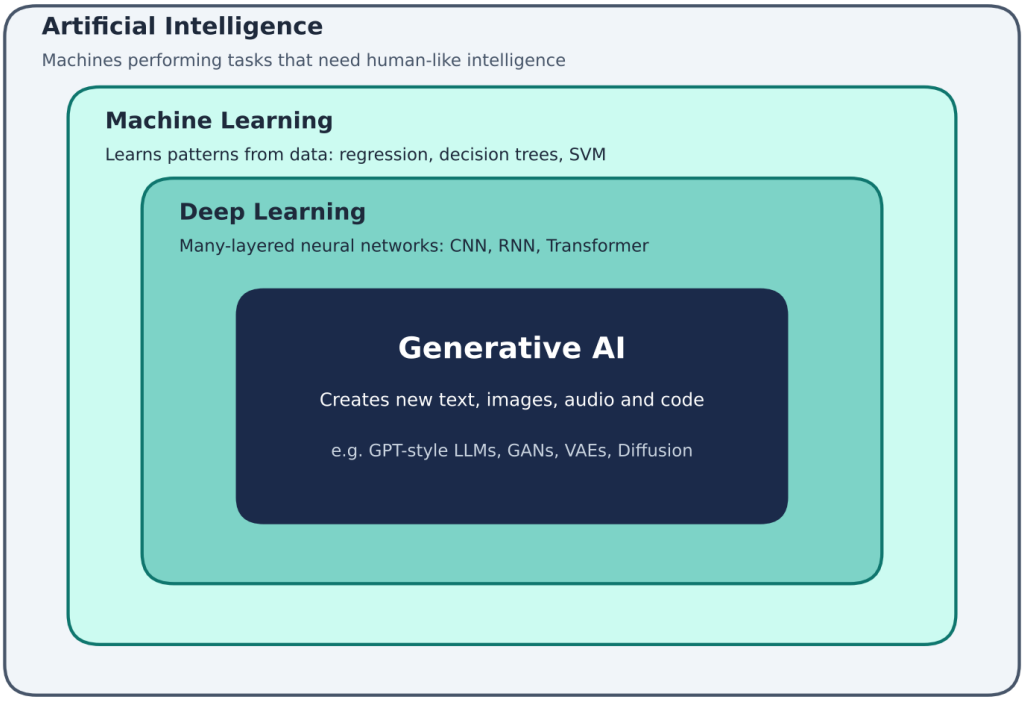


Figure 1. *The hierarchy of AI fields. Each inner circle is a subset of the one around it.*

2.2 Types of machine learning

Paradigm	Learns from	Typical task	Example
Supervised	Labelled examples (input with known output)	Classification, regression	Spam detection, house-price prediction
Unsupervised	Unlabelled data	Clustering, dimensionality reduction	Customer segmentation
Self-supervised	The data itself, by hiding part of it and predicting the hidden part	Pre-training large language models	Predicting the next word in a sentence
Reinforcement	Rewards received after actions in an environment	Control and decision making	Game-playing agents, robotics

KEY NOTE

Most modern language models are trained with **self-supervised learning** during pre-training. They are then refined with supervised fine-tuning and reinforcement learning from human feedback (covered in Chapter 6).

2.3 Discriminative versus generative models

A **discriminative** model learns the boundary between categories, so it can label an input. A **generative** model learns how the data itself is distributed, so it can create new examples that look like the training data. Figure 2 shows the difference.

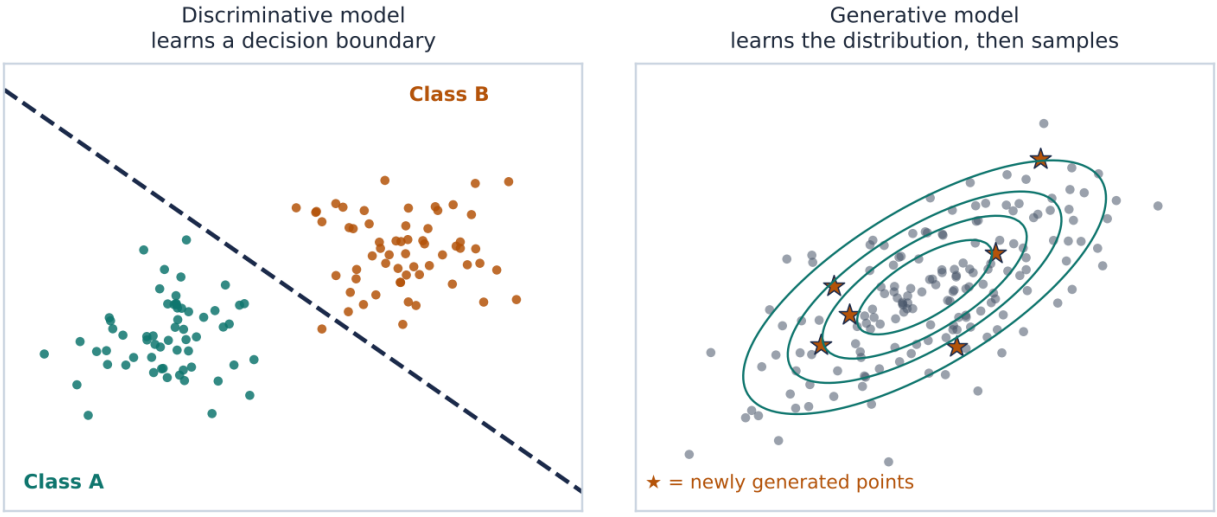


Figure 2. *Left: a discriminative model separates two classes. Right: a generative model learns the shape of the data and draws new points (stars) from it.*

Aspect	Discriminative model	Generative model
Question answered	Given an input x, which class y does it belong to?	What does the data look like, and how can new examples be drawn from it?
Typical output	A label or a number	New samples such as text, images or audio
Examples	Logistic regression, support vector machines, most classifiers	GANs, VAEs, diffusion models, GPT-style language models
Can it create new data?	No	Yes

3 What Is Generative AI?

3.1 Definition

Generative AI refers to models that learn the underlying patterns of a training dataset and use them to produce new, original-looking outputs. The outputs can be text, images, audio, video, code or 3D objects. They resemble the training data without being copies of it.

3.2 How a generative model works

Most generative systems follow the same four stages:

- **Collect data.** A large dataset is gathered, such as billions of words of text or millions of labelled images.
- **Train a model.** A neural network with many parameters (weights) is adjusted so that it assigns high probability to realistic examples.
  - Training repeatedly compares the model's output with real data and corrects the error.
  - The result is a set of learned weights that encode patterns in the data.
- **Run inference.** A user gives a prompt. The model produces an output by sampling from the probabilities it has learned.
- **Refine.** The model is fine-tuned and aligned to behave helpfully and safely (Chapter 6).

3.3 Key terms at a glance

Term	Meaning
Model	A mathematical function, usually a neural network, that maps inputs to outputs.
Parameters (weights)	The numbers the model learns during training. Large models have billions of them.
Training	The process of adjusting parameters so that the model fits the data.
Inference	Using a trained model to generate output for a new input.
Prompt	The input instruction or context that the user gives the model.
Sampling	Choosing an output at random according to probabilities, so the same prompt can produce different results.

> **KEY NOTE**
> Generation is **sampling**. A generative model does not look up a stored answer. It produces its output one piece at a time, choosing each piece from a probability distribution. This is why the same prompt can give different outputs on different runs.

4 Major Families of Generative Models

Four families dominate current generative AI. Each one learns the data distribution in a different way, which gives each a different balance of quality, speed and stability.

4.1 Generative Adversarial Networks (GANs)

GANs were introduced by Ian Goodfellow and colleagues in 2014. A GAN has two neural networks that compete. The **generator** turns random noise into a sample. The **discriminator** tries to tell real samples from generated ones. The generator tries to fool the discriminator, and the discriminator tries to catch the fakes. Figure 3 shows this loop.

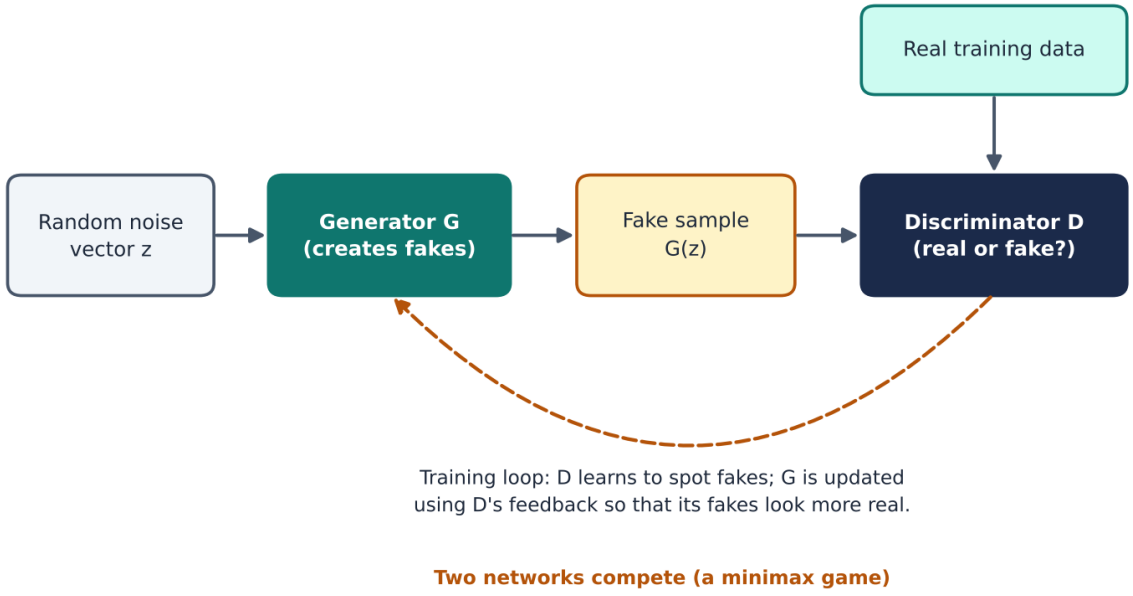


Figure 3. *A GAN. The generator and discriminator are trained against each other.*

- **Strengths:** produces sharp, realistic images and generates a sample in one forward pass.
- **Weaknesses:** training is unstable, and the generator can suffer **mode collapse**, where it produces only a few kinds of output.

4.2 Variational Autoencoders (VAEs)

VAEs were introduced by Kingma and Welling in 2013. An **encoder** compresses an input into a probability distribution in a **latent space**, described by a mean μ and a standard deviation σ. A latent code z is sampled from that distribution, and a **decoder** reconstructs the input from it. The sampling step uses the noise term ε (Figure 4).

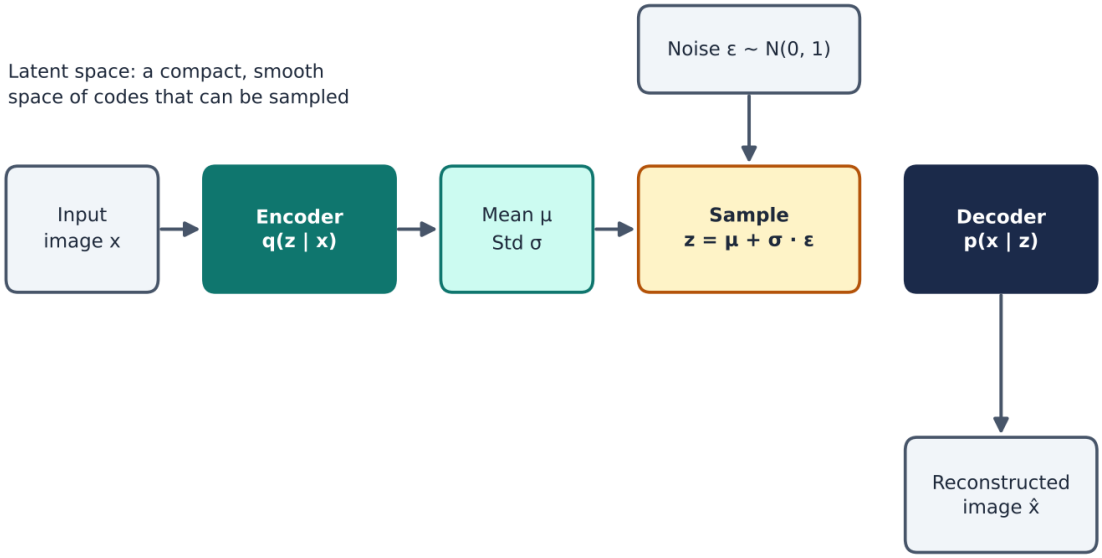


Figure 4. *A variational autoencoder. The sampling step $z = \mu + \sigma \cdot \varepsilon$ keeps training differentiable.*

- **Strengths:** stable training, and a smooth latent space in which nearby codes give similar outputs. This suits interpolation and representation learning.
- **Weaknesses:** reconstructed or generated images are often blurrier than those from GANs or diffusion models.

4.3 Diffusion models

Diffusion models learn to reverse a noising process. In the **forward process**, a little Gaussian noise is added to an image over many steps until only random noise remains. A neural network learns the **reverse process**, predicting and removing the noise one step at a time. To generate something new, the model starts from pure random noise and denoises it step by step (Figure 5). Denoising diffusion probabilistic models were popularised in 2020.

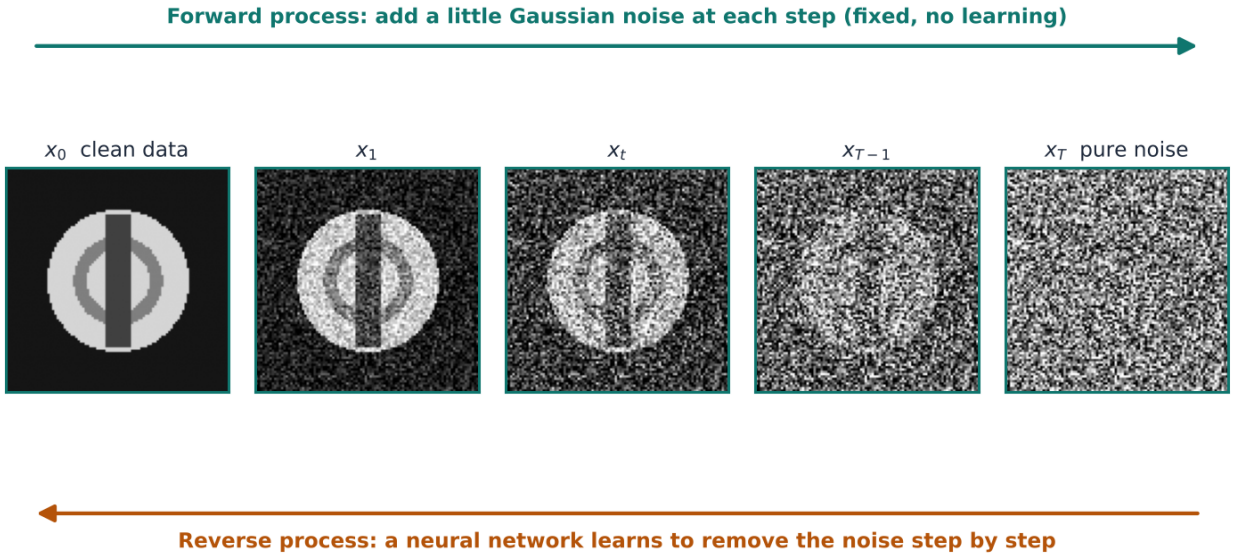


Figure 5. *The forward process adds noise to clean data. The learned reverse process removes it.*

- **Strengths:** high-quality, diverse outputs and stable training. Widely used for image, audio and video generation.

- **Weaknesses:** sampling needs many denoising steps, so generation is slow and compute-heavy.

4.4 Autoregressive transformers (large language models)

An autoregressive model generates a sequence **one token at a time**. Each new token is predicted from all the tokens before it. Large language models (LLMs) are trained on huge amounts of text using next-token prediction, and they are built on the **Transformer** architecture (Chapter 5). Examples include the GPT family, Claude, Gemini and LLaMA.

- **Strengths:** fluent, flexible language skills, usable for many tasks from a single model, and extendable to images and audio.
- **Weaknesses:** can produce confident but false statements (hallucinations), and training and running large models is costly.

4.5 Comparing the families

Family	Core idea	Typical output	Main strength	Main weakness
GAN	Generator and discriminator compete	Images, video	Sharp, fast sampling	Unstable training, mode collapse
VAE	Encode to a latent distribution, then decode	Images, molecules	Stable, smooth latent space	Blurrier outputs
Diffusion	Learn to reverse gradual noising	Images, audio, video	High quality and diversity	Slow sampling, compute-heavy
Autoregressive transformer	Predict the next token	Text, code (and images)	Flexible, strong language skill	Hallucinations, large compute

4.6 Landmark papers

Year	Paper	Contribution
2013	Kingma and Welling, Auto-Encoding Variational Bayes	Introduced the VAE.
2014	Goodfellow et al., Generative Adversarial Nets	Introduced GANs.
2017	Vaswani et al., Attention Is All You Need	Introduced the Transformer.
2020	Ho, Jain and Abbeel, Denoising Diffusion Probabilistic Models	Made diffusion models practical for high-quality image generation.

5 The Transformer Architecture

The Transformer, introduced in 2017, is the architecture behind most modern language models. Its central mechanism is **self-attention**, which lets every token look at the other tokens in the input.

5.1 Tokens and embeddings

A **tokenizer** splits text into **tokens**. A token may be a whole word, a part of a word or a single character. Each token is mapped to an integer ID. Each ID is then mapped to an **embedding**, a list of numbers that represents the token's meaning. Tokens with related meanings receive embeddings that are close together.

5.2 Self-attention

For each token, the model creates three vectors: a **query** (Q), a **key** (K) and a **value** (V). The query of one token is compared with the keys of all tokens to measure how relevant each one is. The values are then combined using those weights. The standard formula is:

$$Attention(Q, K, V) = softmax(QK^T / \sqrt{d_k}) V$$

Here d_k is the size of the key vectors. Dividing by its square root keeps the numbers in a stable range. The softmax turns the scores into weights that add up to 1.

> **KEY NOTE**
> Self-attention is why context matters. The word *bank* gets a different representation in *river bank* and in *bank account*, because attention draws on the surrounding words.

5.3 Other components of a Transformer block

- **Positional encoding:** attention alone ignores word order, so position information is added to each embedding.
- **Multi-head attention:** several attention operations run in parallel, so the model can track different kinds of relationship at once.
- **Feed-forward network:** a small neural network applied to each token separately.
- **Residual connections and layer normalisation:** keep training stable in deep networks.
- **Stacking:** blocks are repeated many times. Large models use dozens to over a hundred blocks.

5.4 How an LLM generates text

Figure 6 shows the loop that produces each response.

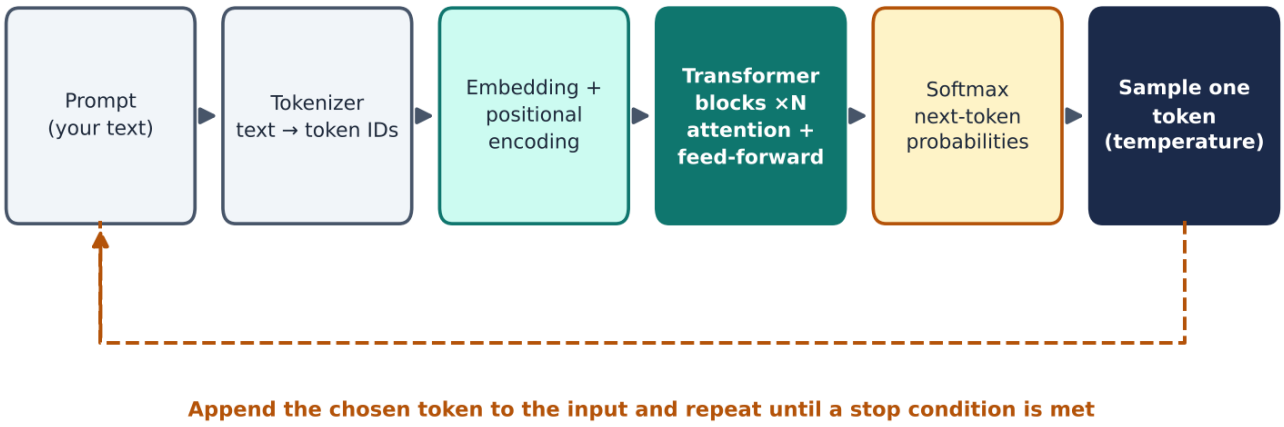


Figure 6. *The generation loop of a transformer language model.*

- **Tokenize** the prompt into token IDs.
- **Embed** the tokens and add positional information.
- **Process** the sequence through the stacked Transformer blocks.
- **Output** a probability for every token in the vocabulary. The softmax produces these probabilities.
- **Sample** one token from that distribution, append it to the input and repeat.

5.5 Temperature and sampling

Temperature controls how the next token is chosen. A low temperature (for example 0.2) makes the model focus on the most likely tokens, so the output is predictable. A high temperature gives more variety and creativity, but it also raises the risk of errors. **Top-k** and **top-p** sampling restrict the choice to a shortlist of the most likely tokens.

> **EXAM TIP**
> Temperature is a trade-off between **creativity** and **reliability**. Tasks that need one correct answer, such as arithmetic or code, usually use a low temperature.

6 Training, Adaptation and Prompting

A model's general ability is shaped in stages. The stages differ in cost, in whether they change the model's weights, and in what they are used for.

6.1 Pre-training

During **pre-training**, the model learns from a very large general corpus using a self-supervised objective, usually next-token prediction. This is the most expensive stage. It can use thousands of GPUs or TPUs for weeks or months.

6.2 Fine-tuning

Fine-tuning continues training on a smaller, targeted dataset, such as example instructions and answers or domain-specific text. It specialises the model's behaviour.

6.3 Alignment with human feedback

In **reinforcement learning from human feedback (RLHF)**, people compare or rank model outputs. A reward model learns these preferences, and the language model is then tuned to produce answers that people judge helpful, honest and safe.

6.4 Prompt engineering

Prompt engineering improves results by changing the input rather than the model. Useful techniques include giving a clear role, stating the required format, supplying a few worked examples (*few-shot* prompting) and asking the model to reason step by step.

6.5 Retrieval-augmented generation (RAG)

In **RAG**, the system searches a document store for passages relevant to the question, then places them in the prompt. The model answers from the retrieved text. RAG reduces hallucination and lets a system use up-to-date or private information without retraining.

6.6 Comparing adaptation methods

Method	Changes model weights?	Relative cost	Typical purpose
Prompt engineering	No	Low	Better answers from the same model
RAG	No	Medium	Answers grounded in specific or current documents
Fine-tuning	Yes	Medium to high	Specialised style, task or domain
Pre-training	Yes	Very high	Building a general-purpose base model

7 Applications

Generative AI is used across most sectors. The table lists representative examples by domain.

Domain	Example applications
Text and communication	Drafting emails, summarising reports, translation, chatbots and virtual assistants
Software development	Code generation and explanation, bug finding, writing unit tests, documentation
Images and design	Text-to-image creation, product mock-ups, editing (inpainting) of photographs
Audio and music	Text-to-speech, voice synthesis, music and sound-effect generation
Video	Text-to-video clips, visual effects, animation of still images
Science and healthcare	Protein and molecule design, drafting candidate compounds, summarising literature (with expert review)
Education	Personalised tutoring, quiz and question generation, revision notes
Business	Marketing copy, customer-support automation, synthetic data for training other models

8 Challenges, Risks and Responsible Use

Generative AI brings real benefits and real risks. Responsible use means naming each risk, recognising it in practice and applying a suitable safeguard.

Risk	What it means	Possible safeguard
Hallucination	Confident but incorrect output	Grounding (RAG), citations, human checking
Bias	Models reflect unfair patterns in their training data	Diverse data, bias testing, review
Misinformation and deepfakes	Realistic fake images, audio or video that mislead people	Content labelling, watermarking, provenance standards
Copyright and ownership	Unclear rules on training data and on who owns the output	Licensed datasets, clear terms of use, legal review
Privacy and security	Sensitive data pasted into prompts or leaked by a system	Do not share confidential data; use approved tools; access control
Environmental cost	High electricity and cooling-water use in training and serving	Efficient models, renewable energy, sensible model sizing
Over-reliance	Using output without understanding it, which undermines learning	Use AI as a tutor; verify and explain the work yourself
Evaluation difficulty	No single metric captures quality	Combine benchmarks with human evaluation

EXAM TIP

When asked about a risk, follow a three-part answer: **name** the risk, give a **concrete example**, and state one **mitigation**.

9 Summary and Glossary

9.1 Summary

- AI is the broad field. Machine learning learns from data. Deep learning uses many-layered neural networks. Generative AI creates new content.
- Discriminative models classify inputs. Generative models learn the data distribution and can sample from it.
- GANs use a generator and a discriminator that compete. VAEs encode inputs into a latent distribution and decode them. Diffusion models learn to remove noise step by step.
- Large language models are autoregressive Transformers. They predict one token at a time, using self-attention to weigh the context.
- Output is produced by sampling. Temperature and top-k or top-p settings control the balance between creativity and reliability.
- Pre-training, fine-tuning, RLHF, prompt engineering and RAG shape model behaviour.
- Applications are broad, and so are the risks. Responsible use means checking outputs and protecting data.

9.2 Glossary

Term	Meaning
Artificial intelligence (AI)	Systems that perform tasks that normally need human intelligence.
Machine learning (ML)	Methods that learn patterns from data instead of explicit rules.
Deep learning	Machine learning with neural networks that have many layers.
Generative AI	Models that create new text, images, audio, code or other content.
Discriminative model	A model that predicts a label or value from an input.
Generative model	A model that learns the data distribution and can produce new samples.
Token	A unit of text (word, part of a word or character) processed by a language model.
Embedding	A vector of numbers that represents the meaning of a token.
Self-attention	A mechanism in which each token weighs the relevance of every other token.
Transformer	A neural architecture built on self-attention; the basis of modern LLMs.
Large language model (LLM)	A large Transformer trained to predict text, used for many tasks.
GAN	Generative adversarial network: a generator and a discriminator trained against each other.
VAE	Variational autoencoder: encodes inputs to a latent distribution and decodes them.
Diffusion model	A model that generates content by reversing a gradual noising process.

Term	Meaning
Latent space	A compressed, learned space in which data is represented by a few numbers.
Hallucination	Fluent output that is factually wrong or unsupported by the input.
Temperature	A setting that controls how random the token choices are during sampling.
Fine-tuning	Further training of a pre-trained model on a smaller, specific dataset.
RAG	Retrieval-augmented generation: adding retrieved documents to the prompt.
Parameters (weights)	The learned numbers that define a model's behaviour.

9.3 Key papers

The four papers in Section 4.6 are the starting point for further study. Read the abstract and introduction of each one first, then the sections that describe the method.